

Fair PN-IQ, GMP and PNNN Comparison

Validated run: 20260714_013938

Objective and methodology

The study compares a complex generalized memory polynomial (GMP), its phase-normalized real I/Q formulations, and phase-normalized neural networks (PNNNs) under one frozen data partition and one temporal complex-NMSE definition. Coupled PN-IQ is an exact real representation of the selected complex GMP; Independent PN-IQ relaxes that I/Q coupling; the sparse PNNN is trained as a wider N12 network and magnitude-pruned to the Independent PN-IQ active-parameter target.

Fair comparison protocol

All models use the same 491,520-sample capture and the local modeled-block X/Y convention (`mappingMode=xy_forward`, without assigning an additional PA-forward meaning). DC is removed once. The deterministic `FIT_POOL` contains 19,661 samples (4.00004%); its disjoint `TEST` complement contains 471,859. `TRAIN` (16,711) and `VAL` (2,950) partition `FIT_POOL`. Support and lambda are selected from `TRAIN/VAL`; linear supports and coefficients are regenerated on `FIT_POOL`. PNNN normalization and gradient updates use `TRAIN`, and `VAL` alone selects dense and fine-tuned checkpoints; PNNN is not refitted on `FIT_POOL`. `TEST` is used once after every final model is frozen. The selected linear lambdas are 0 in this run.

Main results

Model	Active real parameters	Test NMSE (dB)	Dense core FLOPs/sample	Ideal sparse FLOPs/sample	Special ops /sample
Complex GMP DOMP-100	200	−37.94	1,098	1,098	24
Coupled PN-IQ DOMP-100	200	−37.94	1,062	1,062	26
Independent PN-IQ	358	−38.492	1,062	1,062	26
Complex GMP parameter-matched	352	−38.126	1,868	1,868	28
PNNN H4 dense	350	−33.459	876	876	38
PNNN N12 dense	1,046	−36.47	2,252	2,252	54
PNNN N12 sparse	358	−35.457	2,252	876	54

Computational cost

FLOPs refer to inference per modeled output sample; DOMP is used only during identification and is therefore excluded. The core convention assigns one FLOP to a real addition or multiplication, two to a complex addition, and six to a complex multiplication. ELU, worst-case exponential, square-root, division, and absolute-value calls are reported separately because no platform-independent conversion was imposed.

For the sparse N12 PNNN, the two execution views are:

Execution view	Real mult.	Real add.	Core FLOPs/sample	ELU/sample	Special ops/sample
Dense matrices	1,176	1,076	2,252	12	54
Ideal sparse kernel	488	388	876	12	54

The dense count executes the full N12 matrices, including stored zeros. The ideal sparse count includes only active weights and is attainable only if the implementation actually skips zero products; pruning alone does not guarantee this behavior in MATLAB or hardware. The 12 ELUs remain in both views.

Short interpretation

- Independent PN-IQ gives the best TEST NMSE among the highlighted parameter-matched models: -38.492 dB with 358 active real parameters. The closest complex GMP uses 352 parameters and reaches -38.126 dB; sparse N12 uses exactly 358 and reaches -35.457 dB.
- Sparse N12 has a lower *ideal* core count (876) than Independent PN-IQ (1062), but a higher dense-execution count (2252). Its nonlinear activations also add unweighted special operations.
- The coupled and complex DOMP-100 predictions agree to a relative error of 1.320×10^{-11} , confirming a representation change rather than an NMSE improvement.
- No single model wins every metric: PN-IQ leads NMSE here, whereas an implementation-aware sparse PNNN could lead ideal arithmetic count.

Limitations and traceability

Results are from one capture, one neural seed, a 4% fit pool, and analytical operation counts. They do not establish universal accuracy, physical latency, energy, FPGA resources, or hardware superiority. The experiment does not compare quantization, synthesis, or valid demodulated ACPR/EVM. Every number above is read from the validated `comparison_results.csv`, `parameter_summary.csv`, `complexity_flops.csv`, and `selected_hyperparameters.csv`; no experimental model was rerun.